# ASSIGNMENT-4.3


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**BATCH – 46**

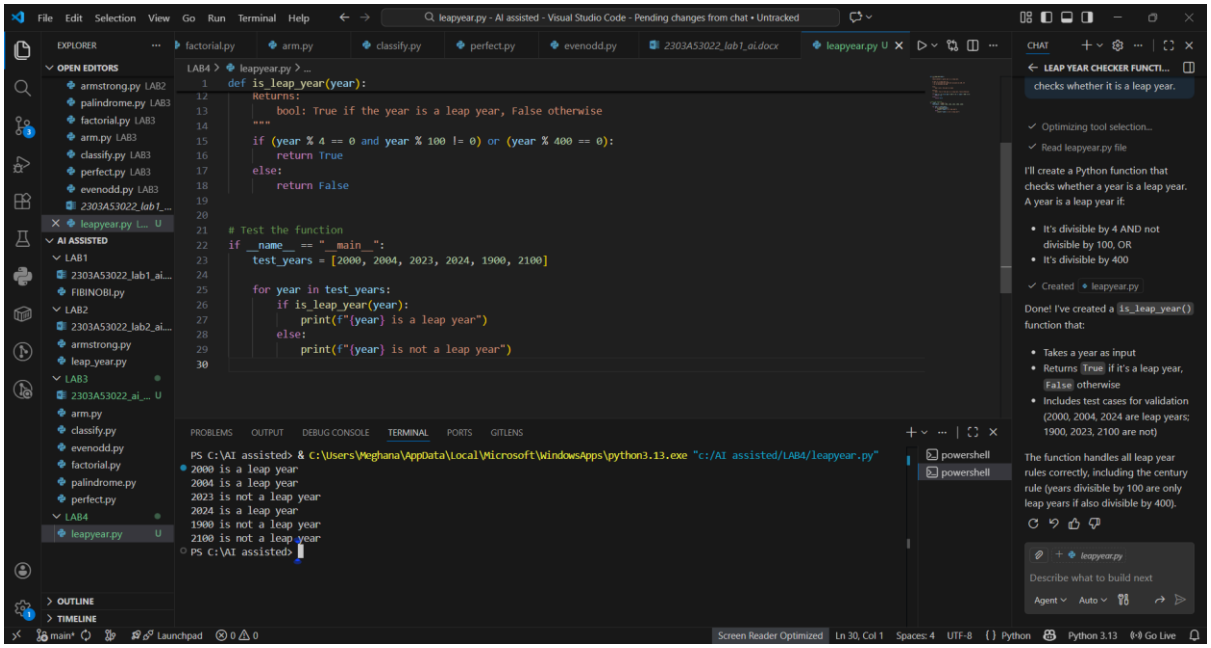

**Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Technique**

**Task 1: Zero-Shot Prompting – Leap Year Check**

**Prompt Used (Zero-Shot)**

Write a Python function that accepts a year as input and checks whether it is a leap year.

**AI-Generated Python Code**




**Explanation**

- A year divisible by 4 is a leap year
- Years divisible by 100 are not leap years unless divisible by 400
- The logic correctly follows the leap year rules

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**Task 2: One-Shot Prompting – Centimeters to Inches Conversion**

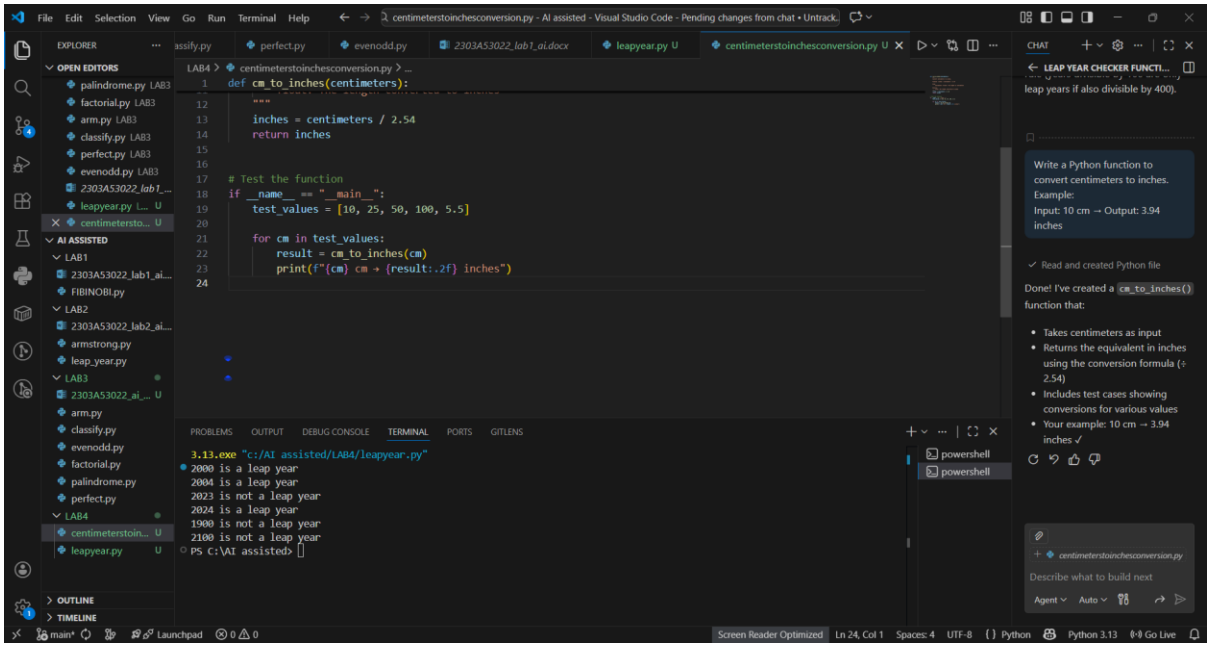
**Prompt Used (One-Shot)**

Write a Python function to convert centimeters to inches.

Example:

Input: 10 cm → Output: 3.94 inches

**AI-Generated Python Code**




**Explanation**

- 1 inch = 2.54 cm
- The function divides centimeters by 2.54
- One example was enough to guide correct logic

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**Task 3: Few-Shot Prompting – Name Formatting**

**Prompt Used (Few-Shot)**

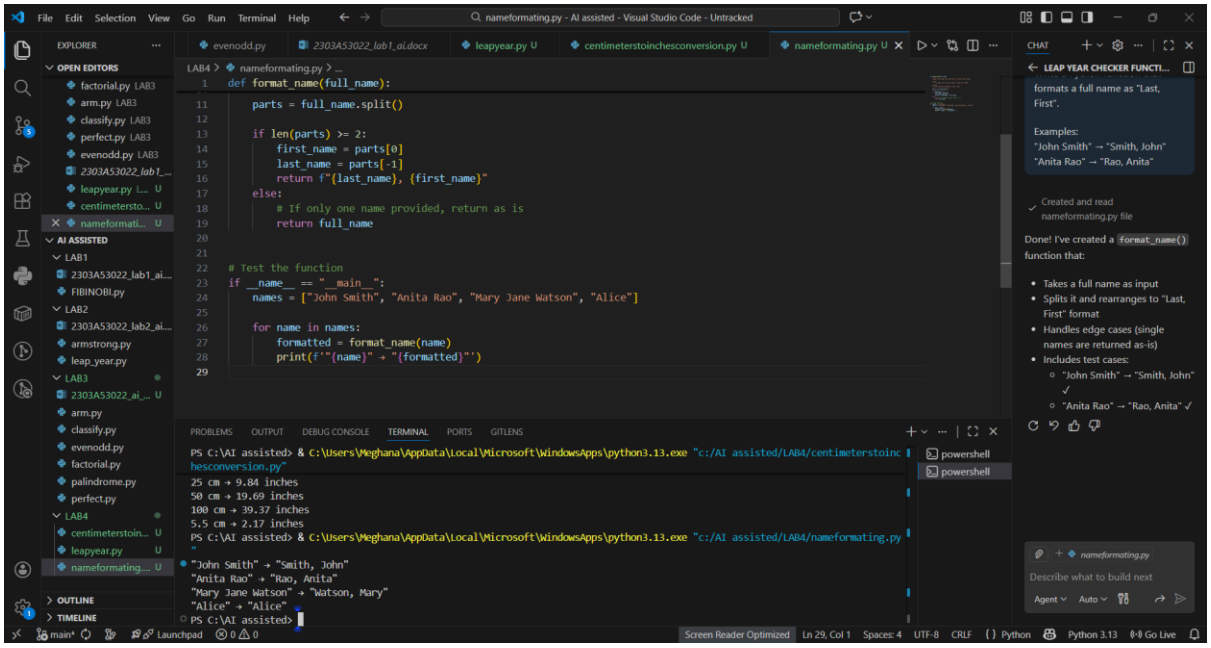
Write a Python function that formats a full name as "Last, First".

Examples:

"John Smith" → "Smith, John"

"Anita Rao" → "Rao, Anita"

**AI-Generated Python Code**




**Explanation**

- Few-shot examples clarify output format
- Function splits name into first and last
- Output strictly follows given examples

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**Task 4: Comparative Analysis – Zero-Shot vs Few-Shot**

**Problem: Count Vowels in a String**

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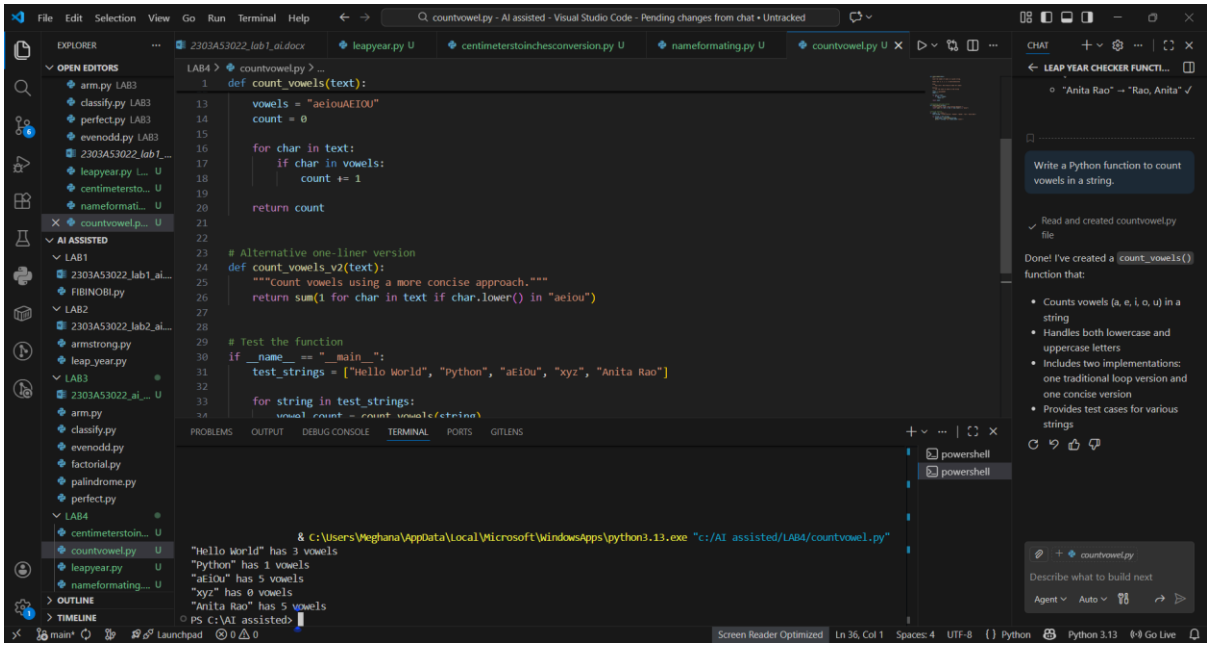
**Zero-Shot Prompt**

Write a Python function to count vowels in a string.

**Zero-Shot Output**




**Few-Shot Prompt**

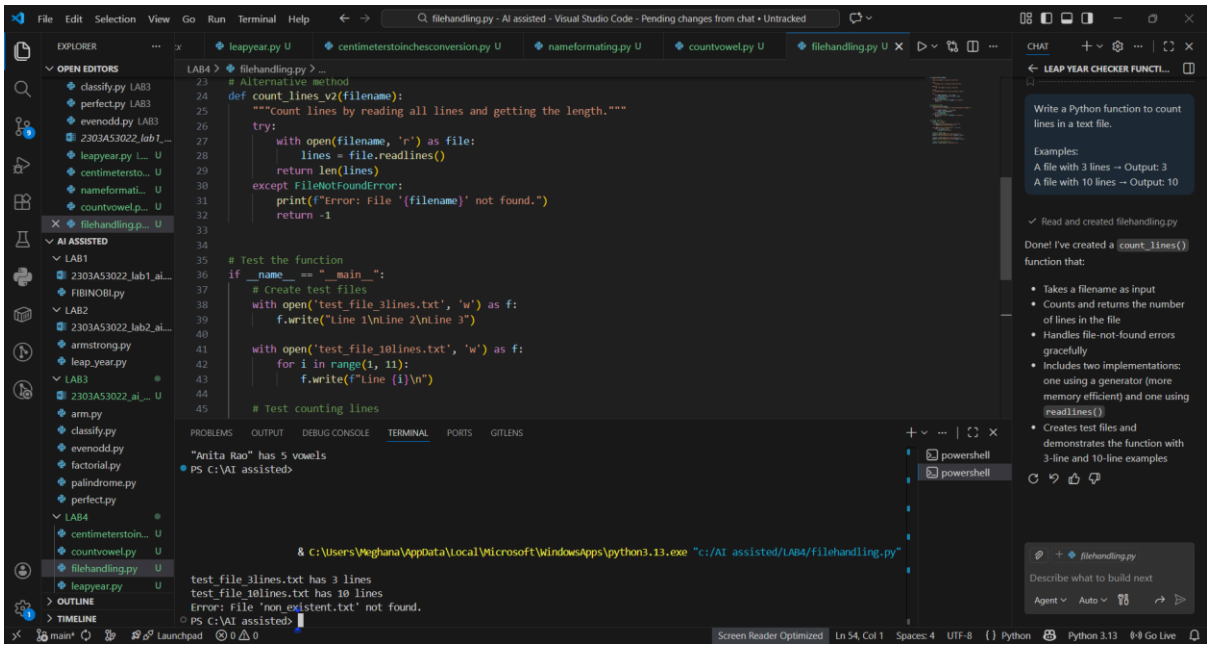
Write a Python function to count vowels in a string.

Examples:

"hello" → 2

"AI Tools" → 4

**Few-Shot Output**

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**Comparison Table**

| Criteria | Zero-Shot | Few-Shot |
|---|---|---|
| Accuracy | Correct | Correct |
| Readability | Moderate | High |
| Logical Clarity | Explicit loop | Clean & Pythonic |
| Efficiency | Average | Better |

**Conclusion**

Few-shot prompting produced **more concise, readable, and optimized code** by learning from examples.

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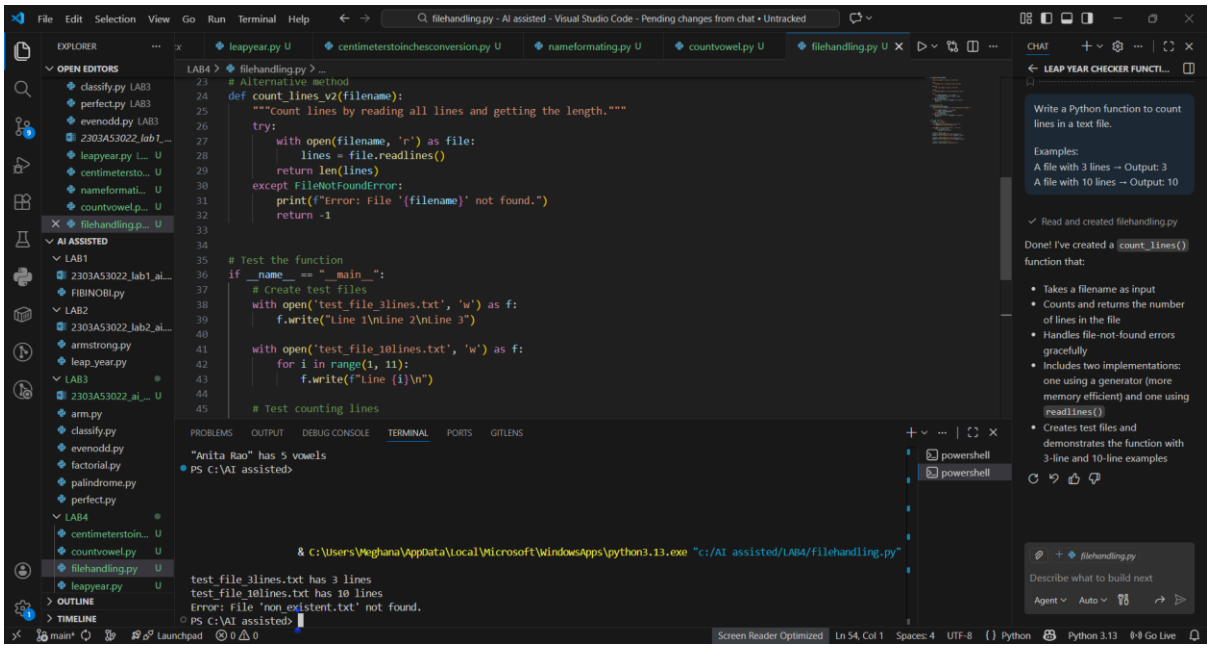
**Task 5: Few-Shot Prompting – File Handling**

**Prompt Used (Few-Shot)**

Write a Python function to count lines in a text file.




Examples:

A file with 3 lines → Output: 3

A file with 10 lines → Output: 10

**AI-Generated Python Code**

**Explanation**

- File opened in read mode
- readlines() returns list of lines

- Length of list equals number of lines

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**Overall Conclusion**

- **Zero-shot** works well for simple, well-known problems
- **One-shot** helps clarify expected behavior
- **Few-shot** produces the best quality code for formatting and logic-heavy tasks
- Providing examples improves accuracy, readability, and confidence in AI outputs